SPECIAL EVENTS

PHASES OF THE MOON

	FULL MOON	NEW MOON
2020	January 10	January 24
2021	January 28	January 13
2022	January 17	January 2
2023	January 6	January 21
2024	January 25	January 11
2025	January 13	January 29
2026	January 3	January 18
2027	January 22	January 7

THE PLANETS

2021: January 24 Mercury is at greatest evening elongation, magnitude –0.3.

2022: January 7 Mercury is at greatest evening elongation, magnitude –0.3.

2023: January 30 Mercury is at greatest morning elongation, magnitude 0.1.

2024: January 12 Mercury is at greatest morning elongation, magnitude 0.0.

2025: January 10 Venus is at greatest evening elongation, magnitude –4.4.

2025: January 16 Mars is at opposition, magnitude –1.4.

2026: January 10 Jupiter is at opposition, magnitude –2.2.

2027: January 3 Venus is at greatest morning elongation, magnitude –4.1.

ECLIPSES

2028: January 11–12 A partial eclipse of the Moon is visible from the Americas, Europe, Africa, and western Asia.

2028: January 26 An annular eclipse of the Sun is visible from northern South America, southwestern Europe, and northwest Africa, with a partial eclipse visible in North and South America, the Atlantic, western Europe, and west Africa.

JANUARY

In both the Northern and Southern Hemispheres, the January evening sky is dominated by the magnificent constellation of Orion, the hunter. He is depicted with raised club and shield, facing Taurus the bull, with his two dogs, Canis Major and Canis Minor, following at his heels. The hazy band of the Milky Way arches from southeast to northwest in northern skies, while in the Southern Hemisphere, the Large Magellanic Cloud lies high up in the sky.

NORTHERN LATITUDES

THE STARS

Sirius, the brightest star in the entire sky, is well displayed on January evenings, twinkling above the southern horizon at mid-northern latitudes. Sirius forms the southern apex of a group of three stars known as the Winter Triangle (see p.436), which is completed by Procyon and Betelgeuse.

Directly overhead for mid-northern observers is the yellowish star Capella, which is the most northerly 1st-magnitude star and the brightest member of Auriga. In the northeast, the Big Dipper stands on its handle and the Square of Pegasus sinks low in the western sky. In the northwest, the Milky Way passes through Auriga into Perseus and Cassiopeia.

DEEP-SKY OBJECTS

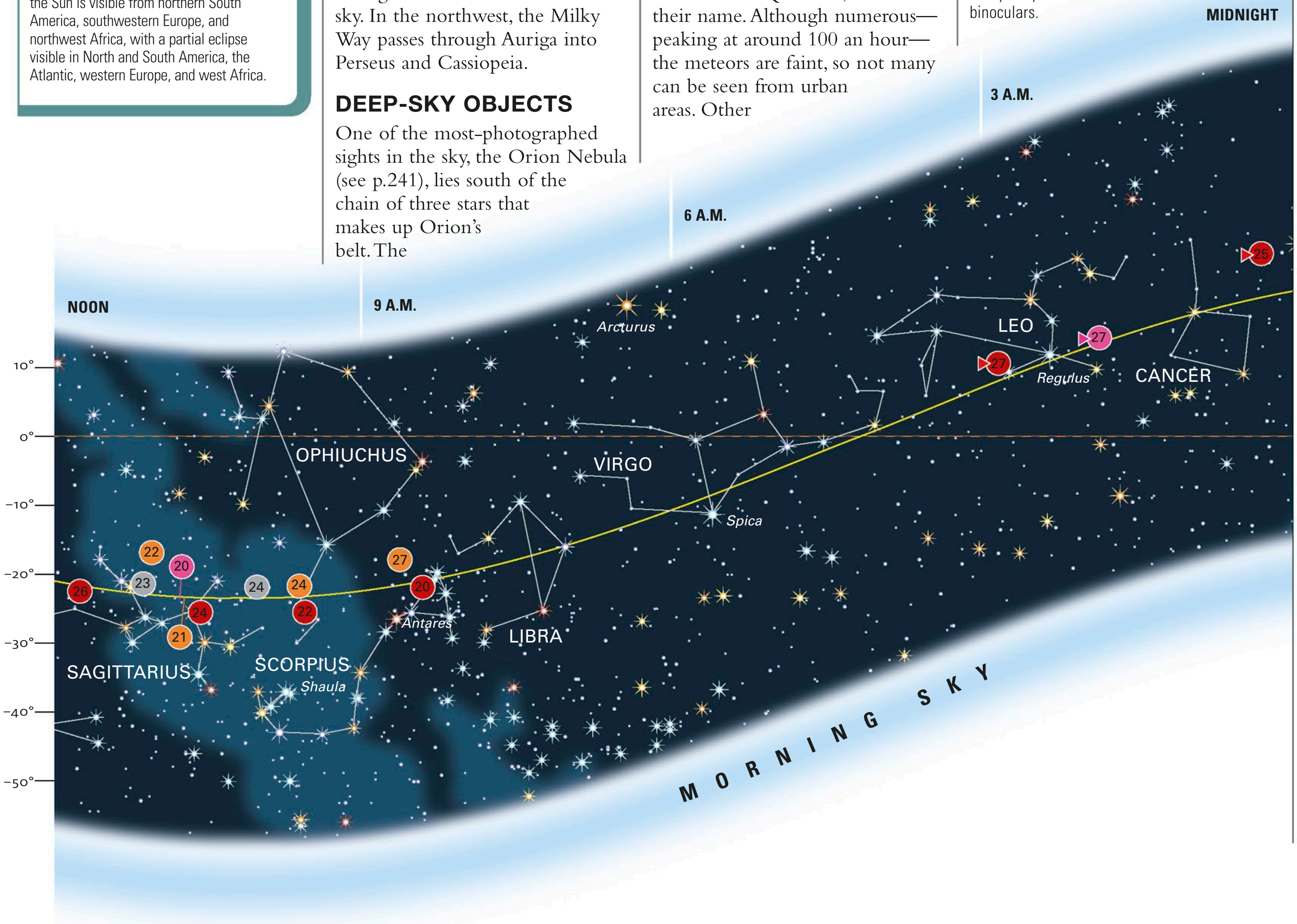
One of the most-photographed sights in the sky, the Orion Nebula (see p.241), lies south of the chain of three stars that makes up Orion's belt. The nebula is easily visible through binoculars from most northern latitudes, and even under average skies, it can be seen with the naked eye as a hazy patch.

Three open star clusters in Auriga—M36, M37, and M38—can be picked out with binoculars.

METEOR SHOWERS

Northern observers can observe the Quadrantid meteors around January 3–4 every year. The meteors radiate from a point near the handle of the Big Dipper in Ursa Major, an area which was once occupied by the now-obsolete constellation Quadrans, hence their name. Although numerous—peaking at around 100 an hour—the meteors are faint, so not many can be seen from urban areas. Other drawbacks are that their peak is short, lasting only a few hours, and their radiant remains low in the northeastern sky until well after midnight.

OPEN CLUSTERS
M36 (center), M37 (left), and M38 (right) in Auriga can be picked out from the Milky Way with binoculars.



THE NIGHT SKY